Literature Status: Twisted Inclusion and Alternating 2-Cocycles

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Status

This is a lightweight `literature_already_answered` packet. It records that the future-work application announced in Choi, *A twisted inclusion between tensor products of operator spaces*, arXiv:1606.06287v2, was written up in the separate later paper Choi, *Constructing alternating 2-cocycles on Fourier algebras*, arXiv:2008.02226v4.

Source Future-Work Item

The source note arXiv:1606.06287 proves that for operator spaces V and W, if $\widetilde{W}$ denotes the opposite operator-space structure on the underlying Banach space of W, then the identity map on $V \otimes W$ extends to a contraction

$$V \widehat{\otimes} W \longrightarrow V \otimes_{\min} \widetilde{W}.$$

This is Theorem `t:mainthm` in the parsed source.

The abstract says that this theorem will be applied in future work to construct antisymmetric 2-cocycles on certain Fourier algebras. The author comments added on 20 May 2020 say that the cocycle application was removed from the historical note and would be written separately in fuller detail.

Supporting Answer

The supporting paper arXiv:2008.02226 is exactly such a separate write-up. Its abstract states that it gives the first examples of locally compact groups whose Fourier algebras support non-zero alternating 2-cocycles, and it lists the twisted inclusion theorem for operator-space tensor products as one of the two main technical ingredients. The arXiv metadata for arXiv:2008.02226v4 also states that the paper includes an improved version of material from arXiv:1606.06287v2.

The theorem-level evidence is explicit. Theorem `t:headline` proves that for $n \geq 4$ and G equal to $\mathrm{SU}(n)$, $\mathrm{SL}(n, \mathbb{R})$, or $\mathrm{Isom}(\mathbb{R}^n)$, there is a non-zero continuous alternating 2-cocycle on the Fourier algebra $A(G)$. Theorem `t:twisted inclusion` restates/proves the twisted inclusion. Theorem `t:mainthm` constructs non-zero alternating 2-cocycles from non-zero co-completely bounded derivations, and Theorem `t:underyournoses` supplies the co-cb derivations needed for the headline examples.

Scope

This is not a new proof. It records the already-known later answer to the future-work application announced in arXiv:1606.06287: applying the twisted inclusion theorem to construct antisymmetric, equivalently alternating, 2-cocycles on certain Fourier algebras. New follow-up questions posed in the final section of arXiv:2008.02226, such as the small-dimensional cases, are not settled by this status packet.

References

- Y. Choi, *A twisted inclusion between tensor products of operator spaces*, arXiv:1606.06287v2.
- Y. Choi, *Constructing alternating 2-cocycles on Fourier algebras*, arXiv:2008.02226v4.